

## RESEARCH INTERESTS

Visual world models, self-supervised visual learning, 3D vision, vision-language models, and diagnostic evaluation. I study how visual systems can learn predictive representations from time, geometry, and action, and how to evaluate whether those representations support reasoning and interaction beyond benchmark shortcuts.

## EDUCATION

- 4/2025 – 3/2027 (expected) **M.Eng. in Computer Science** University of Tsukuba, Japan  
 Advisor: [Prof. Yutaka Satoh](#). GPA: **4.27/4.30**.
- 4/2021 – 3/2025 **B.Eng. in Applied Mechanics and Aerospace Engineering** Waseda University, Japan  
 Advisor: [Prof. Kiyoshi Saito](#). GPA: **3.29/4.00**. Minor in Computer Science. Undergraduate research on reinforcement learning for energy-system optimization.

## SELECTED PUBLICATIONS AND MANUSCRIPTS

- Kohsuke Ide**, Ryosuke Yamada, Yue Qiu, Yoshihiro Fukuhara, Hirokatsu Kataoka, Yuki M. Asano, and Yutaka Satoh.  
***Beyond Downstream Scores: Controlled Diagnostics for Point-Cloud Self-Supervised Learning.***  
 Under review at **NeurIPS 2026**. Reviewer scores: **5/5/5**.
- Kohsuke Ide**, Ryosuke Yamada, Yue Qiu, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh.  
***Structure First, Appearance Later: Label-Efficient Pretraining for Neural Radiance Fields.***  
 Under review at the International Conference on 3D Vision (**3DV**) 2027.
- Kohsuke Ide**, Ryosuke Yamada, Yue Qiu, Xianzheng Ma, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh.  
***Beyond Single Object: Learning 3D Relations with Large Language Models.***  
 IEEE/CVF Conference on Computer Vision and Pattern Recognition Findings (**CVPR Findings**), 2026.  
[project](#) | [paper](#) | [code](#)
- Ryosuke Yamada, **Kohsuke Ide**, Yoshihiro Fukuhara, Hirokatsu Kataoka, Gilles Puy, Andrei Bursuc, and Yuki M. Asano.  
***3D sans 3D Scans: Scalable Pre-training from Video-Generated Point Clouds.***  
 IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.  
[project](#) | [paper](#) | [code](#)
- Hirokatsu Kataoka, Yoshihiro Fukuhara, Yonglong Tian, Shangzhe Wu, Oishi Deb, Ryosuke Yamada, Christian Rupprecht, Jianyuan Wang, **Kohsuke Ide**, Koichi Namekata, Xianzheng Ma, Yiming Chen, Robert Geirhos, Aditi Raghunathan, Yuki M. Asano, Deva Ramanan, David Fouhey, Andrew J. Davison, Yilun Du, Jiajun Wu, and Zhuang Liu.  
***Visual General Intelligence: A White Paper.***  
 arXiv preprint arXiv:2608.25924, 2026. [paper](#)
- Kohsuke Ide**, Ryosuke Yamada, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh.  
***Seeing Red, Thinking Bad: Color Bias in Vision Language Models.***  
 International Conference on Pattern Recognition (**ICPR**), 2026.  
[project](#) | [paper](#) | [code](#)
- Kohsuke Ide**, Ryosuke Yamada, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh.  
***Colors You Can't See, Semantic Biases You Can't Ignore.***  
 ICCV Workshop on Multimodal Reasoning and General Intelligence (**ICCV Workshop**), 2025.

## RESEARCH EXPERIENCE

- 9/2025 – Present **Research Assistant, AIST** Tsukuba, Japan
- Conduct research on visual world models, 3D understanding, vision-language models, and diagnostic evaluation.
  - Lead projects from problem formulation and implementation through experiments and writing; this work has produced first-author papers at **CVPR Findings 2026** and **ICPR 2026**.
  - Collaborate with researchers at Oxford's Visual Geometry Group on 3D and video representation learning.

- 1/2024 – 8/2025 **Research Intern, AIST** Tsukuba, Japan
- Developed diagnostic evaluations for shortcut learning, language leakage, and visual biases in 3D and vision-language models.
  - Built multi-object 3D reasoning benchmarks that require differences between scenes rather than language-only cues.
  - Designed and implemented a video-to-point-cloud data-generation pipeline for scalable 3D pre-training.

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## RESEARCH EXPERIENCE (CONTINUED)

- 8/2024 – 3/2025 **Research Intern, Preferred Networks** Tokyo, Japan
- Developed and improved [3D reconstruction and free-viewpoint video](#) pipelines.
  - Optimized rendering and model components for efficient experimentation.
- 9/2021 – 3/2024 **Machine Learning Engineer Intern, Lightblue Technology** Tokyo, Japan
- Developed and deployed an object-tracking system for a work-efficiency monitoring application using Docker and AWS EC2.
  - Improved human-action-recognition accuracy by 20% through feature engineering and [Optuna](#)-based hyperparameter search.
- 9/2023 – 10/2023 **MLOps Engineer Intern, M3** Tokyo, Japan
- Built [RecoverOOM](#), an open-source, event-driven service for automatically recovering GKE CronJobs after out-of-memory failures.
  - Implemented OOM detection through Cloud Logging and Pub/Sub, automatic memory-limit updates, and immediate job re-runs.

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## RESEARCH VISITS & INTERNATIONAL COLLABORATION

- 1/19 – 1/31/2026 **Research Visit** University of Oxford, UK  
Research meetings with members of the Visual Geometry Group and Active Vision Laboratory.
- 4/22 – 4/30/2026 **ASPIRE Computer Vision Workshop and Research Visit** University of Oxford, UK  
Presented research and discussed follow-up work with researchers at Oxford VGG.

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## ACADEMIC SERVICE

- **Workshop Organizer**, [Visual General Intelligence: Vision Research Toward the AGI Era](#), CVPR 2026.

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## HONORS & AWARDS

- **MIRU 2025 Interactive Presentation Award** (Jul 2025)  
Selected as one of **11** award recipients from **601** interactive presentations (**top 1.8%**) for "Can 3D Large Language Models Count to Three?"
- **WINC Hackathon Winner** (2023)  
Built a ResNet-based push-up counter optimized with TensorRT for Jetson Nano.

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## TECHNICAL SKILLS & LANGUAGES

- **Research:** Computer vision, visual world models, self-supervised learning, 3D reconstruction, vision-language models, diagnostic evaluation.
- **Programming and systems:** Python, C++, Go, PyTorch, CUDA, TensorRT, OpenCV, Git, Docker, Kubernetes, Linux, AWS, and GCP.
- **Languages:** Japanese (native); English (fluent; TOEFL iBT 118/120, TOEIC 990/990).